Runyi Yang
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EDUCATION

INSAIT, Sofia Bulgaria(ETHz & TU Darmstadt) **Feb 2025 - Present**
PhD Researcher in Computer Vision and Robotics, advised Prof. Luc Van Gool
Research Topic: Spatial Understanding, Robotics and Physical AI

Imperial College London, London, United Kingdom **Sep 2023 - Sep 2024**
Master of Research (MRes) in AI and Machine Learning, Computing, advised by Dr. Tolga Birdal
Research Topic: Quantifying Camera Relocalization Uncertainty via Conformal Prediction

Beijing Institute of Technology (BIT), Beijing, China **Sep 2019 - Jun 2023**
Bachelor of Engineering in Automation, GPA: 3.8/4.0, Score: 90.1/100, Rank: 5%
Awards: Outstanding Graduates of BIT, Outstanding Student Award, Academic Excellence Scholarship × 8, Student Representative of School of Automation
Interested Fields: Scene Understanding; 3D Reconstruction; Real2Sim2Real

EXPERIENCE

Litchi AI (Startup in Robotic World Model) **Hybrid**
Research Scientist Intern *Apr 2026 - Present*
- Robot Learning Infra build up from scratch.
- Physical AI and World Action Model Training.

Noetix Robotics **Hybrid**
Robotics Intern *Jan 2026 – Mar 2026*
- Robot Learning Infra for dexterous manipulation.
- VLA training for dexterous hands tool use.

Pixomondo Innovation Lab, Sony **London, Hybrid**
Research Scientist *May 2024 – Feb 2025*
- Integrate Integrating advanced 3D reconstruction methods, e.g. NeRFs and 3D Gaussian Splatting to VFX.
- Enable language-embeded 3DGS and use it for object removal and inpainting.
- Accommodate the current tools to viser as a viewer.

DISCOVER Lab, Institute for AI Industry Research, Tsinghua University **Hybrid**
Research Intern, Advised by Prof. Zhou Guyue and Prof. Zhao Hao. *May 2022 – April 2024*
- Research in **Neural Radiance Fields** and **3D reconstruction**. Improve the sampling methods of point clouds for the implicit representation. Integrate graph-based methods to NeRFs and 3D Gaussian Splatting. Achieved balanced performance of rendering effects like PSNR, storage and performance.
- Research in **camera pose estimation** (corresponding to SLAM), use deep learning methods to generate a robust loop-closure module. Build a loop-closure system for Xiaomi Robots indoor use.

Data Study Group, The Alan Turing Institute **London, UK**
Facilitator *Nov 2023 - Dec 2023*
- Use LSTM, NARX and Transformer to forecast time series control variables in chemical processing scenarios. Achieved less than 0.1 RMSE in the prediction. https://doi.org/10.5281/zenodo.13847810
- Use conformal prediction methods to quantify the prediction region and uncertainty of the forecasting model.

Selected PUBLICATIONS & Pre-Prints | **Bold: Highlight,** Underline: With links.

- **UniPhyArt: Physics-aware Articulated Object Generation from a Single Image with a Multi-Modality Large Language Model.** (In Submission)

- InsideArt: Articulated 3D Generation with Interior Synthesis and Object Placement from a Single Image. (In Submission)
- ***ϕ*-RIE: From Photorealistic Reconstruction to Interactive Environments.** (In Submission)
- **Do LiDAR Language Models Really Understand Spatio-temporal Relationships?** (In Submission)
- LangStreet: Persistent Language Fields for Anchor-Decoded Street Gaussians. (In Submission)
- ConeGaussian: Anti-Aliased Gaussian Ray-Tracing for Generic Central Cameras. (In Submission)
- **DexNinja: Learning Robust Dexterous Cutting Policy with a Real-to-Sim-to-Real Data Engine. (CORL 2026, ICRA R2S2R Workshop Hightlight)**
- CoRef-GS: Cooperative Referring Gaussian Splatting for Multi-Agent Scene Understanding. (In Submission)
- GeoStereo: A Unified Stereo Geometry Estimation Framework for Disparity and Surface Normal (ACM MM 2027)
- HeatSurf: Thermal Cues Enable Temperature-Aware robotic manipulation. (In submission)
- **Chorus: A Multi-Teacher Pretraining Framework for Holistic 3D Gaussian Scene Encoding. (CVPR 2026 Award List & Oral)**
- **Rethinking Expressivity and Degradation-Awareness in Attention for All-in-One Blind Image Restoration. (ICLR 2026)**
- EgoNight: Towards Egocentric Vision Understanding at Night with a Challenging Benchmark. (ICLR 2026, *arxiv 2510.06218*)
- SceneSplat++: A Large Dataset and Comprehensive Benchmark for Language Gaussian Splatting. (NeurIPS 2025, *arxiv 2506.08710*)
- **SceneSplat: Gaussian Splatting-based Scene Understanding with Vision-Language Pretraining. (ICCV 2025 Oral, *arxiv 2503.18052*)**
- CityLoc: 6 DoF Localization of Text Descriptions in Large-Scale Scenes with Gaussian Representation. (In submission, *arxiv 2501.08982*)
- **SUNDAE: Spectrally Pruned Gaussian Fields with Neural Compensation. (In submission, arxiv 2405.00676)**
- GaussianGrasper: 3D Language Gaussian Splatting for Robotic Grasping. (IEEE RA-L, arxiv 2403.09637)
- **City-scale continual neural semantic mapping with three-layer sampling and panoptic representation. Knowledge-based Systems.**
- **MARS: An Instance-aware, Modular and Realistic Simulator for Autonomous Driving. CAAI International Conference on Artificial Intelligence. (Best Paper Runner-up Award, 2/348)**

HONORS & PRIZES

- ICML Silver Reviewer — May 2026
- CVPR 2026 Oral & Award List — May 2026
- ICCV 2025 Oral — Sep 2025
- CICAI 2023 Best Paper Runner-up Award. — May 2023

SKILLS and LANGUAGE and Activities

- Programming: **Python (PyTorch)**, **C & C++**, **MATLAB**, Web (PHP+JavaScript+html5), CUDA C
- IELTS: 7.0 (Listening 7.5, Reading 8.0, Writing 7.0, Speaking 6.0), CET-6: 557/710
- *Founder, President*, BIT Swimming Club — Dec 2019 - Jun 2022
- *Captain*, BIT Swimming Team — Mar 2021 - Jun 2022